UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450
EXAMINATION	:	NOVEMBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the ICT450 mock paper (November 2026). Answers are unofficial model answers, not official UTM marking schemes.

2. Marking points state the expected answer and the allocation of marks; every question retains its source paper and page reference.

3. Five questions x 20 marks; total = 100 marks.

4. Real-bank questions: 5. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 12 unique ICT450 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 8 printed pages.

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MARKING GUIDE

QUESTION 1 (20 marks)

Source: ICT450-2018-DEC · page [2] · bank id ICT450-2018-DEC-Q1

Reference answer

A database system environment is made up of the people, procedures, data, hardware and software that work together to manage and use a database.

(a) Four components of a database system environment. [8 marks]

- Hardware — the physical devices on which the database runs, including servers, storage, network equipment and user terminals.
- Software — the DBMS, application programs, utilities and development tools that let users define, access and manage the data.
- People — the database administrator, designers, analysts, programmers and end users who use or manage the database.
- Data — the raw facts stored in the database, together with the metadata that describes their structure and meaning.
- Procedures — the written instructions and policies for using, backing up, securing and recovering the database.

(b) Four characteristics of successful Internet-age databases. [8 marks]

- Ubiquitous connectivity — users can reach the database from web browsers, mobile apps and data services at any time and from any location.
- Scalability and performance — the system can handle growing volumes of users, transactions and data without unacceptable response times.
- High availability and reliability — the database remains accessible through failures, with replication, backup and recovery facilities.
- Security and privacy — authentication, authorisation, encryption and auditing protect data that is exposed over public networks.
- Manageability — the database can be monitored, tuned and administered centrally with minimal disruption.

(c) Two advantages and two disadvantages of an ERD. [4 marks]

- Advantage: an ERD gives designers, programmers and users a clear visual model of the entities, attributes and relationships before implementation.
- Advantage: it is independent of any DBMS, so the same conceptual design can be reviewed and implemented on different platforms.
- Disadvantage: a large ERD can become crowded and difficult to read, especially when many relationships and weak entities are involved.
- Disadvantage: an ERD does not show every business rule, process or physical implementation detail, so it must be supported by other documentation.

Marking guide

- a) 2 marks for each of four components named and explained (4 × 2 = 8).
- b) 2 marks for each of four characteristics named and explained (4 × 2 = 8).
- c) 1 mark for each advantage and each disadvantage, correctly explained (4 × 1 = 4).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.

Knowledge base basis: TOPIC 1 DATABASE CONCEPT (1).pdf; TOPIC 2 DATA MODEL (1).pdf; TOPIC 3 RELATIONAL DATA MODELstu.pdf

Assumptions / notes

- The question asks for four components; five are listed so the answer remains complete if the marker expects a different set.

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QUESTION 2 (20 marks)

Source: ICT450-2018-JUN · page [3] · bank id ICT450-2018-JUN-Q2

Reference answer

(a) Database life cycle phases. [8 marks]

- i) Implementation and loading: the physical database is created from the design — tables, keys, constraints, views, indexes and security are defined; existing data is converted and loaded; backup and recovery arrangements are put in place and the database is tested before release.
- ii) Three backup levels:
- Full backup — a complete copy of the whole database; it is the baseline for recovery but takes the most time and space.
- Differential backup — copies only data changed since the last full backup; recovery needs the last full backup plus the latest differential backup.
- Transaction-log backup — copies the transaction log, allowing point-in-time recovery and recovery of changes made after the last full or differential backup.
- Incremental backup may be accepted in place of differential backup: it copies changes since the last backup of any type.

(b) Big data and three technologies for managing it. [8 marks]

- Big data is data that is too large, fast-changing or varied for traditional relational systems to capture and process within an acceptable time. Its core characteristics are volume, velocity and variety; veracity and value are often added as extended characteristics.
- Hadoop and HDFS — Hadoop is a Java-based, open-source, fault-tolerant distributed framework; HDFS is its distributed file system for storing very large files across a cluster.
- MapReduce — an open-source programming API that processes large datasets in parallel and provides fast data-analytic services.
- NoSQL — a large-scale distributed database system that stores structured and unstructured data efficiently, with high scalability, availability and fault tolerance.

(c) Structured versus unstructured data. [4 marks]

- Structured data has a predefined schema and is stored in rows and columns with consistent data types; it is easy to query with SQL. Example: a student table containing StudID, Name and CGPA.
- Unstructured data has no predefined relational schema and cannot be stored naturally in rows and columns. Example: images, video, audio, free-text documents or social-media posts.

Marking guide

- a)(i) 2 marks for a correct description of implementation and loading.
- a)(ii) 2 marks for each of three backup levels correctly described (3 × 2 = 6).
- b) 2 marks for the definition of big data; 2 marks for each of three technologies (2 + 3 × 2 = 8).
- c) 2 marks for structured data with an example and 2 marks for unstructured data with an example (2 + 2 = 4).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 2 DATA MODEL (1).pdf; TOPIC 6 DATABASE DESIGN (1).pdf

Assumptions / notes

- The three backup levels are interpreted as full, differential and transaction-log backup, the standard DBLC recovery model.

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QUESTION 3 (20 marks)

Source: ICT450-2019-DEC · page [4] · bank id ICT450-2019-DEC-Q4

Reference answer

The invoice in Figure 3.0 contains customer, invoice and item data in one form. The item lines repeat for the same invoice, so the form is first flattened to a one-row-per-item relation in 1NF.

(a) Relational schema for the order form. [2 marks]

Order-form relation in 1NF (one row per item line).

```
INVOICE_ITEM (InvoiceNo, InvoiceDate, CustomerNo, CustomerName, CustomerAddress, ItemNo, ItemName,
Model, Quantity, ItemPrice, LineTotal, TotalAmount)
```

Invoice header data

InvoiceNo	InvoiceDate	CustomerNo	CustomerName	Address	TotalAmount
2088	26-Aug-2019	153	Puan Adriana Yusuf	Taman Rasah Jaya, Seremban	902.95

Item lines

InvoiceNo	ItemNo	ItemName	Model	Qty	ItemPrice	LineTotal
2088	CRB101	Electric Kids Car	Buggati	1	559.00	559.00
2088	LEG101	City Fire Ladder Truck	LEGO	1	184.00	184.00
2088	LEG102	City ATV Race Team	LEGO	2	79.99	159.98

(b) Primary key of the relation. [2 marks]

The primary key is the composite (InvoiceNo, ItemNo). InvoiceNo alone is not unique because one invoice has many items, and ItemNo alone is not unique because the same item can appear on many invoices.

(c) Normalise the relation to 3NF. [16 marks]

Step 1 — First normal form (1NF): the relation in (a) already stores one item line per row and all values are atomic. Its key is (InvoiceNo, ItemNo).

1NF — one row per invoice item.

```
INVOICE_ITEM_1NF (InvoiceNo, InvoiceDate, CustomerNo, CustomerName, CustomerAddress, ItemNo, ItemName,
Model, Quantity, ItemPrice, LineTotal, TotalAmount)
```

Step 2 — Second normal form (2NF): remove partial dependencies on part of the composite key.

2NF — invoice, item and invoice-item relations.

```
INVOICE (InvoiceNo, InvoiceDate, CustomerNo, CustomerName, CustomerAddress, TotalAmount)
ITEM (ItemNo, ItemName, Model, ItemPrice)
INVOICE_ITEM (InvoiceNo, ItemNo, Quantity, LineTotal)
```

Step 3 — Third normal form (3NF): remove transitive dependencies. CustomerNo determines CustomerName and CustomerAddress, so customer data is moved to CUSTOMER.

3NF — customer, invoice, item and invoice-item relations.

```
CUSTOMER (CustomerNo, CustomerName, CustomerAddress)
INVOICE (InvoiceNo, InvoiceDate, CustomerNo, TotalAmount)
ITEM (ItemNo, ItemName, Model, ItemPrice)
INVOICE_ITEM (InvoiceNo, ItemNo, Quantity, LineTotal)
```

3NF sample data: CUSTOMER

CustomerNo	CustomerName	CustomerAddress
153	Puan Adriana Yusuf	Taman Rasah Jaya, Seremban

3NF sample data: INVOICE

InvoiceNo	InvoiceDate	CustomerNo	TotalAmount
2088	26-Aug-2019	153	902.95

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3NF sample data: ITEM

ItemNo	ItemName	Model	ItemPrice
CRB101	Electric Kids Car	Buggati	559.00
LEG101	City Fire Ladder Truck	LEGO	184.00
LEG102	City ATV Race Team	LEGO	79.99

3NF sample data: INVOICE_ITEM

InvoiceNo	ItemNo	Quantity	LineTotal
2088	CRB101	1	559.00
2088	LEG101	1	184.00
2088	LEG102	2	159.98

Marking guide

- a) 2 marks: a relation containing the invoice, customer and repeating item data.
- b) 2 marks: composite key (InvoiceNo, ItemNo) with a reason.
- c) 16 marks: 1NF item relation stated with atomic rows (4), 2NF partial dependencies removed into INVOICE, ITEM and INVOICE_ITEM (6), 3NF transitive dependency removed into CUSTOMER (6).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 5 NORMALIZATION (1).pdf

Assumptions / notes

- LineTotal is Quantity × ItemPrice; TotalAmount is the sum of LineTotal for the invoice.
- The invoice is treated as one order form with a repeating group of item lines.

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QUESTION 4 (20 marks)

Source: ICT450-2019-JUN · page [4] · bank id ICT450-2019-JUN-Q3

Reference answer

Relational schema used by the question.

```
BRAND (brandID, brandName)
BOOKING (bookID, equipID*, studID*, bookDate, returnDate)
STUDENT (studID, studName, studAddr, studPhone)
EQUIPMENT (equipID, equipName, equipPrice, category, brandID*, quantity, supplier)
```

(a) Create the BOOKING table. [3 marks]

```
CREATE TABLE BOOKING (
    bookID      VARCHAR(10) NOT NULL,
    equipID     VARCHAR(10) NOT NULL,
    studID      VARCHAR(10) NOT NULL,
    bookDate    DATE,
    returnDate  DATE,
    PRIMARY KEY (bookID),
    FOREIGN KEY (equipID) REFERENCES EQUIPMENT (equipID),
    FOREIGN KEY (studID)  REFERENCES STUDENT (studID)
);
```

(b) Increase equipment prices by 10%. [3 marks]

```
UPDATE EQUIPMENT
SET equipPrice = equipPrice * 1.10;
```

(c) Change the width of the student address column. [3 marks]

```
ALTER TABLE STUDENT
MODIFY studAddr VARCHAR(50);
```

For SQL Server, the equivalent statement is ALTER TABLE STUDENT ALTER COLUMN studAddr VARCHAR(50).

(d) Equipments supplied by Al-Ikhsan in the racket or ball category, sorted by category. [3 marks]

```
SELECT *
FROM EQUIPMENT
WHERE supplier = 'Al-Ikhsan'
  AND category IN ('racket', 'ball')
ORDER BY category ASC;
```

(e) Equipments that have not been booked by any student. [4 marks]

```
SELECT e.*
FROM EQUIPMENT e
WHERE NOT EXISTS (
    SELECT 1
    FROM BOOKING b
    WHERE b.equipID = e.equipID
);
```

A LEFT JOIN with IS NULL is also correct: SELECT e.* FROM EQUIPMENT e LEFT JOIN BOOKING b ON e.equipID = b.equipID WHERE b.equipID IS NULL.

(f) Number of equipment types under each brand; show only brands with more than five types. [4 marks]

```
SELECT brandID,
       COUNT(DISTINCT equipID) AS number_of_types
FROM EQUIPMENT
GROUP BY brandID
HAVING COUNT(DISTINCT equipID) > 5;
```

Marking guide

- a) 3 marks: CREATE TABLE with columns (1), primary key (1) and both foreign keys (1).
- b) 3 marks: correct UPDATE (2) and 10% calculation (1).
- c) 3 marks: correct ALTER TABLE (2) and VARCHAR(50) (1).

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- d) 3 marks: supplier and category predicates (2), ORDER BY category (1).
- e) 4 marks: correct anti-join/not-exists logic (3), correct equipment output (1).
- f) 4 marks: GROUP BY brand (1), COUNT (1), HAVING > 5 (2).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 7 SQL.pdf

Assumptions / notes

- The printed BOOKING schema is missing a comma between studID* and bookDate; the answer inserts the comma.
- Data types and sizes are not printed, so reasonable types are chosen and stated in the CREATE TABLE answer.
- The SQL is written in portable SQL; Oracle/MySQL may also accept MODIFY instead of ALTER COLUMN.

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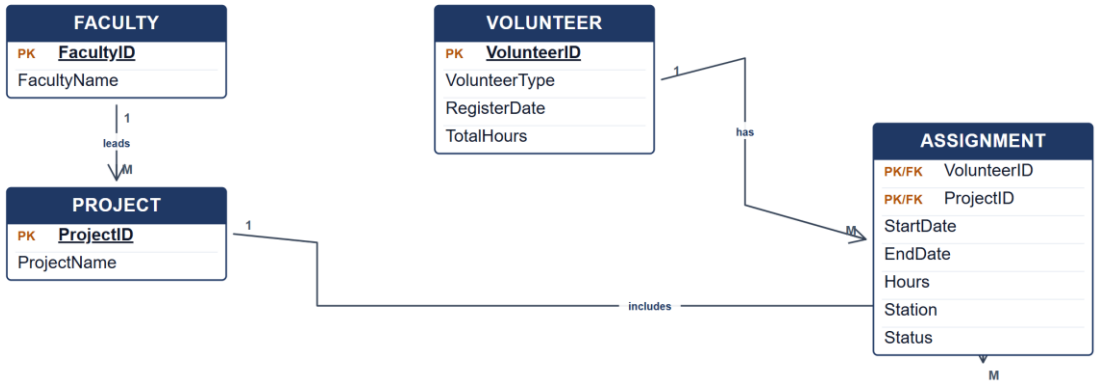
QUESTION 5 (20 marks)

Source: ICT450-2022-FEB · page [6] · bank id ICT450-2022-FEB-Q5

Reference answer

The ERD treats a volunteer as one entity that may be a student or a staff member. The many-to-many participation between volunteers and projects becomes the ASSIGNMENT associative entity, which carries the assignment details. Each project is led by one faculty.

(a) Construct the ERD with Crow's Foot notation. [16 marks]



Crow's Foot ERD for the volunteer management system.

(b) Two reports from the ERD. [4 marks]

- Volunteer hours by project and faculty — join FACULTY, PROJECT and ASSIGNMENT to total Hours for each project, with the leading faculty shown.
- Volunteer participation report — list each volunteer with the projects, stations, start/end dates and status, so coordinators can see who is assigned where.
- Project status report — count assignments and total hours per project, grouped by status.

Marking guide

- a) 16 marks: VOLUNTEER (2), PROJECT (2), FACULTY (2), ASSIGNMENT with its five attributes (4), correct 1:N relationships and Crow's Foot cardinalities (4), primary and foreign keys (2).
- b) 2 marks for each of two relevant reports explained (2 × 2 = 4).

Provenance: model_unofficial · confidence medium · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 4 ERD_stu (2).pdf; TOPIC 3 RELATIONAL DATA MODELstu.pdf

Assumptions / notes

- A volunteer is recorded once with VolunteerType (Student or Staff); student- and staff-specific attributes are outside the given requirements.
- The project assignment is an associative entity because it has its own attributes: start date, end date, hours, station and status.

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